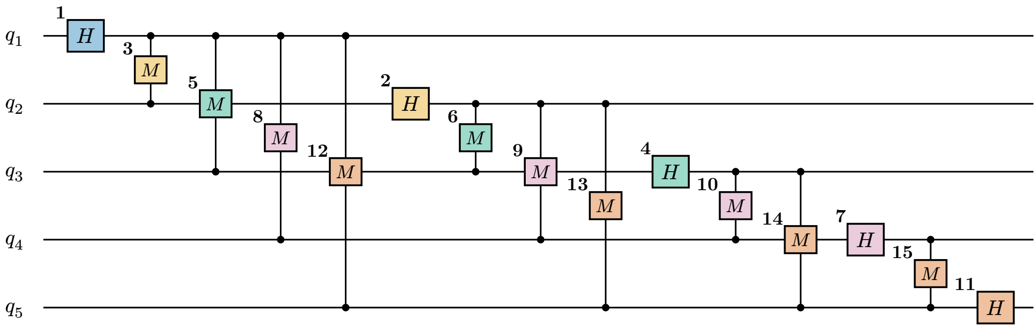
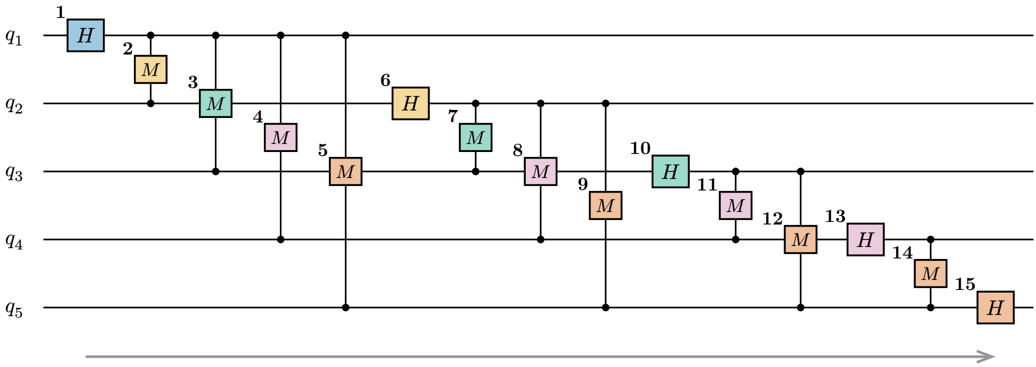
QFT gate-unfreezing — three orderings (single axis, m = 5)

Same QFT(5) circuit in the paper's tensor-network notation (controlled-phase M = box between two wires with legs to a dot on each). Bold badge = **unfreeze step** for that ordering; fill = block-stage **k** (= highest qubit a gate touches). Cumulative: each step thaws one more gate, the rest stay frozen at identity. (No SWAP gates — bit-reversal is folded into the index convention.)

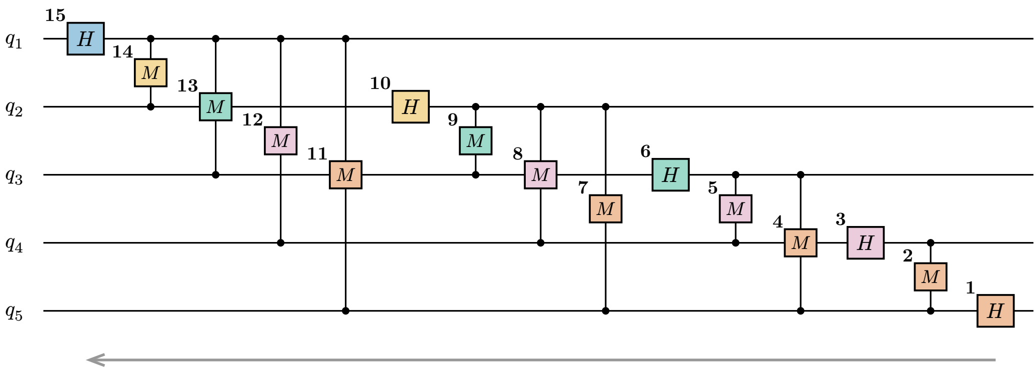
(a) block-growth stage k completes QFT(k, k); long-range couplings delayed — the gate-level mirror of the block sweep



(b) left → right sweep unfreeze in QFT construction order 1→15; q1 couples to all of q2–q5 before H_2 at step 6



(c) right → left sweep reverse construction order; last-built gate (H_5) first, H_1 last



Block-stage colours: k = 1 k = 2 k = 3 k = 4 k = 5